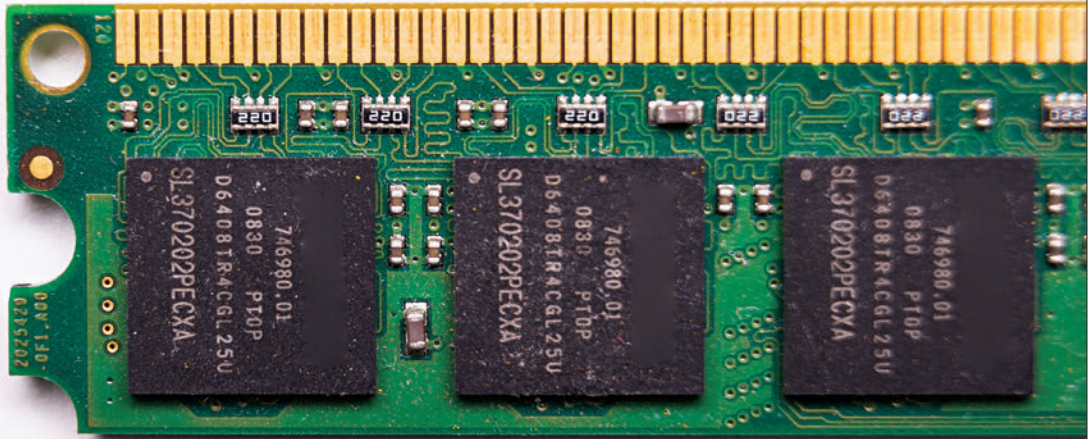




Memory Matters – RAM

Modern DDR5 memory unlocking your PC's performance

Upgrading or building a PC in 2025 means thinking beyond the CPU or GPU. Memory (random-access memory or RAM) is the computer's short-term workspace, holding the data your processor actively manipulates. The quality and quantity of RAM directly affect how responsive your system feels, how many browser tabs you can keep open, and whether your gaming session stutters. This chapter demystifies the complex world of DDR5 modules, explains why they're a giant leap over DDR4, and provides clear guidelines for selecting RAM so you can extract every last drop of performance from your DIY machine.

What does RAM do?

RAM temporarily stores data that your CPU needs right now. When you open an application, the operating system moves program files from the slower solid-state drive (SSD) into RAM for lightning-fast access. More RAM allows your system to hold larger or more applications simultaneously, reducing the need to swap data back to disk. Speed (measured in mega-transfers per second or MT/s) and latency (measured in clock cycles) determine how quickly the CPU reads and writes this data. Modern workloads like AI image generation, 3D rendering and open-world gaming have huge memory footprints, making the right choice of RAM essential.